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Python3

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Count Matching Subsequences

Difficulty: **Medium**Accuracy: **57.65%**Submissions: **51K+**Points: **4**

Given two strings, **s1** and **s2**, count the number of subsequences of string **s1** equal to string **s2**.

Return the total count modulo **1e9+7**.

Examples :

Input: s1 = "geeksforgeeks", s2 = "gks"

Output: 4

Explanation: We can pick characters from s1 as a subsequence from indices [0, 3, 4], [0, 3, 12], [0, 11, 12] and [8, 11, 12]. So total 4 subsequences of s1 that are equal to s2.

Input: s1 = "problemoftheday", s2 = "geek"

Output: 0

Explanation: No subsequence of string s1 is equal to string s2.

Constraints:

$1 \leq s1.size(), s2.size() \leq 10^3$

```
1 class Solution:
2     def countWays(self, s1, s2):
3         MOD = 1000000007
4
5         n = len(s1)
6         m = len(s2)
7
8         dp = [0] * (m + 1)
9         dp[0] = 1
10
11        for i in range(n):
12            for j in range(m - 1, -1, -1):
13                if s1[i] == s2[j]:
14                    dp[j + 1] = (dp[j + 1] + dp[j]) % MOD
15
16        return dp[m]
```